# ECG Dataset Comparison

*A simple report for the research supervisor and AI team*

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Report date 16 October 2026

## What this study tells us

The datasets share some useful features, but we have not yet confirmed any pair as ready to combine for training. This study shows what is similar, what is different, and what still needs checking.

The research goal is to classify an ECG as acute STEMI, NSTEMI, or non-MI. My role in this stage is to assess the data before the AI team develops the models. The study compared published information about eleven ECG datasets. [1, 2]

### The main result

Comparing every dataset with every other dataset gives 55 different pairs. The analysis estimates the preparation effort indicated by the information currently available.

| Preparation level | Pairs | What it means |
|---|---|---|
| Easy or Low | 3 | Few differences were found in the available information. |
| Medium or Moderate | 42 | Conversion or further technical review is indicated. |
| Hard or High | 10 | Larger differences need more careful handling. |

Table 1. Easy, Medium and Hard correspond to the original soft, medium and hard categories. These are study rules for preparation effort, not measurements of data quality. Source: [2].

**All 55 pairs are unconfirmed.** Important information is missing. An Easy rating does not mean a pair is ready for training, and a Hard rating does not mean the data must be discarded.

### What was checked

The run compared documented dataset characteristics. It did not open ECG signal arrays or inspect the downloaded dataset files. Optional checks of local metadata and file headers were skipped. No AI model was trained. [1, 2]

**Recommended next step:** Check available metadata and headers for PTB-XL and MIMIC, then review whether their diagnosis labels support the research task.

## What we compared

Metadata means information about a recording. It describes the signal without examining every electrical measurement. The registry is the reference table where these details and their sources are recorded.

**Sampling rate:** The number of measurements per second per lead. At 500 Hz, each lead has 500 samples each second. Hz is the unit of sampling rate, not a separate property.

**Leads:** Different views of the electrical activity. The names and order of the leads matter, as well as their number.

**Duration:** The length of a recording. A 10-second recording at 500 Hz contains 5,000 samples per lead.

| Dataset | Rate in Hz | Channels | Duration in seconds |
|---|---|---|---|
| PTB-XL | 500 | 12 | 10 |
| MIMIC-IV-ECG | 500 | 12 | 10 |
| Ningbo | 500 | 12 | 10 |
| Chapman-Shaoxing | 500 | 12 | 10 |
| Georgia | 500 | 12 | 5 to 10 |
| CPSC 2018 | 500 | 12 | 6 to 144 |
| CPSC-Extra | 500 | 12 | 6 to 144 |
| SPH | 500 | 12 | 10 to 60 |
| PTB Diagnostic | 1,000 | 15 | Not established here |
| INCART | 257 | 12 | 1,800 |
| CODE-15% | 400 | 12 | 7 to 10 |

Table 2. Values describe the selected releases, not measurements from local files. SPH means Shandong Provincial Hospital. PTB-XL uses the records500 version. CPSC duration refers to a combined source description. CODE also contains padding, so stored array length is not the same as observed recording length. Source: [2].

### Other differences still matter

The study also considered equipment, collection years, amplitude units, calibration, filtering and file formats. Calibration explains how stored numbers become physical measurements. Filtering is processing applied to the signal. These details are not fully documented for every source.

For example, the registry lists PTB-XL collection in 1989 to 1996 and MIMIC in 2008 to 2019. These periods describe the sources; they do not by themselves determine whether the data can be combined. [2]

## Which combinations need more preparation

Choose one dataset on the left and another along the bottom. The square where they meet shows the preparation level. The question mark means the result still needs confirmation.

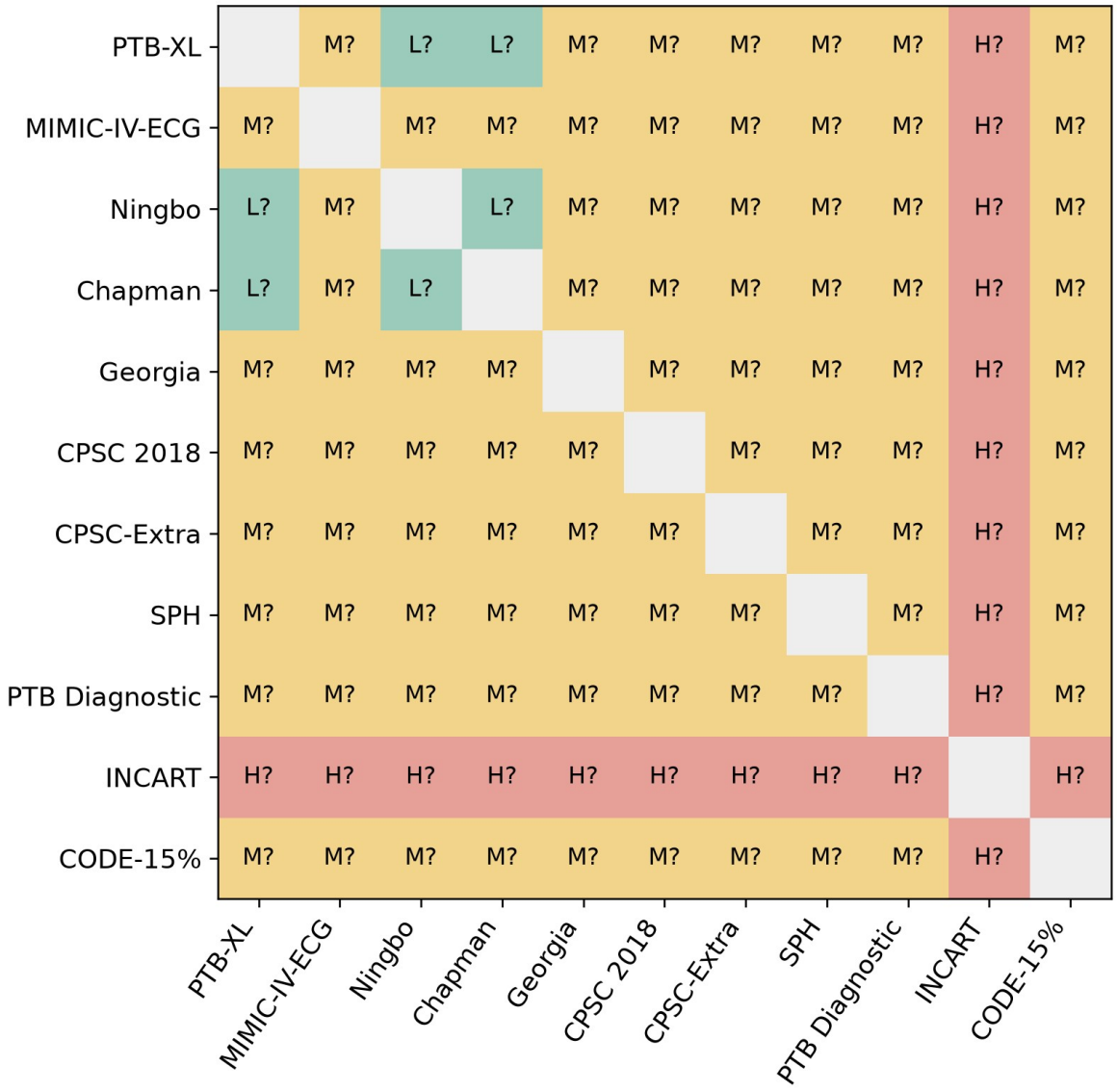



Figure 1. L? = Easy or Low; M? = Medium or Moderate; H? = Hard or High. Chapman means Chapman-Shaoxing. Gray diagonal squares are not assessed. The two halves repeat the same comparisons. Source: [2].

**The three Easy pairs:** PTB-XL with Ningbo, PTB-XL with Chapman-Shaoxing, and Ningbo with Chapman-Shaoxing. Basic similarities do not prove the same equipment, settings or labels.

**The ten Hard pairs:** All involve INCART. Its Holter recordings cover 30 minutes of ECG monitoring, a different recording context from the diagnostic ECG sources. That difference requires a separate decision about its intended use.

## The example of PTB XL and MIMIC

PTB-XL and MIMIC are a practical starting point for further checking because data from both are already being collected for this project. The current assessment rates their pair Medium, with missing information. This is a proposal for further review, not approval to combine them.

### What already matches in the documentation

| Property | PTB-XL | MIMIC-IV-ECG |
|---|---|---|
| Sampling rate | 500 Hz | 500 Hz |
| Lead count | 12 | 12 |
| Recording length | 10 seconds | 10 seconds |

Table 3. Shared basic dimensions in the selected profiles. These values still need checking against the files used by the project. Source: [2].

### What still needs checking

**Lead names and order:** Confirm that each channel represents the expected lead and appears in the agreed order.

**Amplitude units and calibration:** Confirm how to convert stored numbers to the same electrical units. The registry has unresolved MIMIC information here; that does not mean the source files lack it.

**Equipment and processing:** Review available device information and any filtering or preprocessing. Equal sampling rates do not establish equal equipment or settings.

**Diagnosis labels:** Check whether the available evidence supports acute STEMI, NSTEMI and non-MI labels. A general MI annotation does not automatically provide the acute subtype required by this project.

### How much information is missing

For this pair, 8 of the 16 selected technical attributes are known for both sources. That is 50% evidence coverage. It is not 50% model accuracy or 50% download completion. [2]

Across all eleven datasets, 195 of 418 registry entries were documented and 223 were unknown. Unknown means the value was not established in this evidence set. It may be found through further source review or file-header checks. [2]

**Decision for now:** Investigate this pair further. Keep technical checks and clinical-label review as separate requirements before pooled training.

## What the team should do next

The proposed first stage uses metadata and file headers, which are the small descriptions attached to recordings. It does not require loading every waveform. Record how files are selected so a small check is not mistaken for a full dataset audit.

**1. Check the files:** The data analyst should compare available PTB-XL and MIMIC metadata and headers with the documented values. Save a table of observations, missing information and conflicts.

**2. Define the model input:** The AI developer should propose the lead order, electrical units, sampling rate and recording length. The current analysis uses a target of 12 leads, 500 Hz and 10 seconds; this is a study setting, not proof of the best input.

**3. Review the labels:** The data analyst and clinical supervisor should agree how acute STEMI, NSTEMI and non-MI labels will be established. Missing MI annotations should not automatically become confirmed non-MI labels.

**4. Plan the training and test groups:** The analyst and developer should identify repeated patients and overlapping records. Where patient identifiers allow it, keep one patient in one split to reduce misleading evaluation results.

**5. Record the decision:** For each source, document whether it is suitable for the intended training task, needs further checks, or should be left out of that task, with a reason. Keep original files and record any conversions.

### What to give the AI developer

Provide this report, the original results ZIP, the observed metadata table, and the reviewed label definitions when available. Ask the developer to return a proposed input specification and the remaining data requirements.

### A short explanation for the professor

“I compared how eleven ECG datasets were recorded and stored. Some basic features match, but important information is still missing. I classified pairs by the preparation effort indicated by known differences: Easy, Medium or Hard. These are preliminary ratings. My next step is to verify available metadata and headers, then review labels with the clinical supervisor before recommending data for training.”

## How to read the clustering graph

This graph groups datasets using four features: sampling rate, channel count, file format, and diagnostic or Holter recording context. It provides an extra view of similarity.

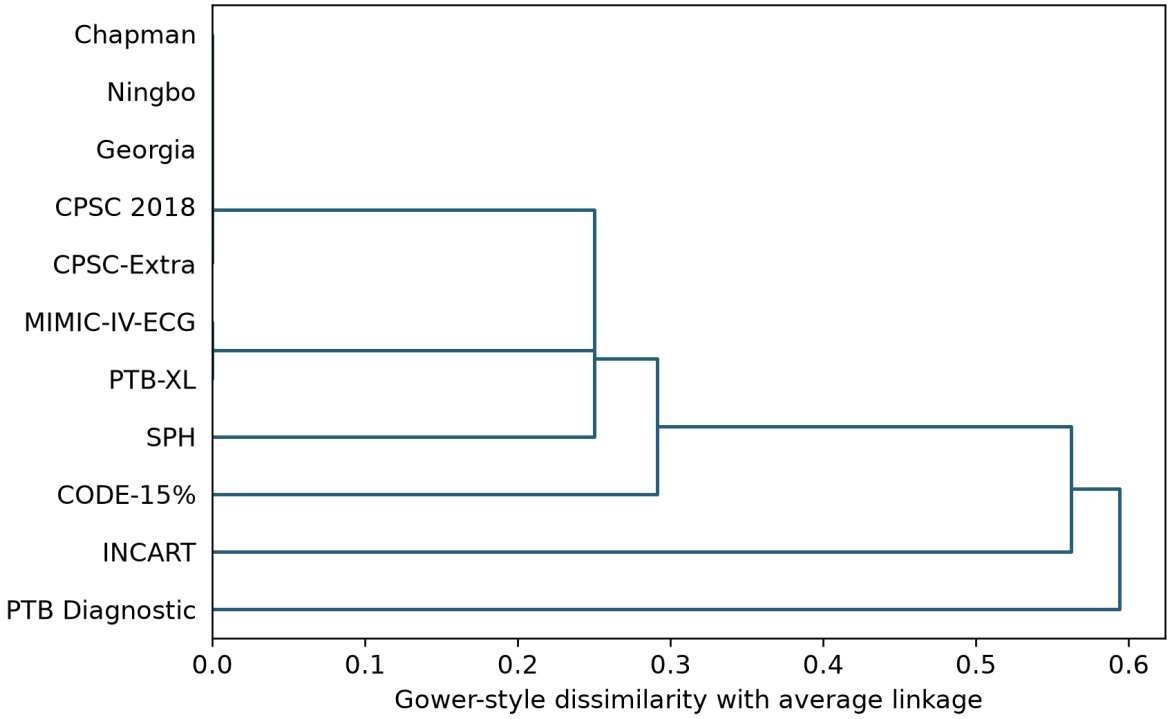



Figure 2. The horizontal axis is a calculated difference score. Datasets whose branches join nearer zero are more similar on the four selected features. Chapman means Chapman-Shaoxing. Source: [2].

**What zero means:** Some datasets match on all four features, so they join at zero. They may still differ in equipment, filters, patients or diagnosis labels.

**How to use this graph:** Use it to describe structural similarity alongside the comparison table. It does not replace missing-information checks or establish readiness for training.

**Where the findings come from**

[1] MI_Metadata_Compatibility_Colab.ipynb - Colab.pdf. Executed notebook export from 10 October 2026. The export records that optional local checks were skipped.

[2] metadata_compatibility_v1.zip. Results from 10 October 2026. Key files: pairwise_assessment.csv, dataset_profiles.csv, attribute_evidence.csv, registry_used.json, linkage_core_equal.csv and run_provenance.json. The registry retains the publisher source references.

For detailed methods, release information and additional evidence tables, use the accompanying ECG Dataset Compatibility Assessment report and the original ZIP.